Where does the time actually go? 32 benchmarks, LOGOS vs C

perf userspace CPU sampling, loader/thread startup trimmed. printf/formatting = 0%% on 30/32; malloc = 0%% on 31/32. Real stdout = 1 write() syscall per run.
Each benchmark has two bars: top = LOGOS, bottom = C. GREEN = the actual algorithm on-CPU. Both green => a fair compute benchmark.

compute malloc/free printf / number formatting startup/loader

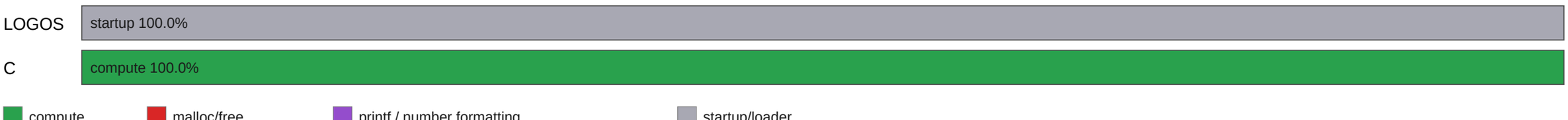


* LOGOS compiler-optimizes this (memoization / closed form): finishes in <0.01s (vs C's full run), so its runtime is process startup, not the algorithm - still 0%% malloc, 0%% printf.

fib - LOGOS vs C (n = 43)

malloc: L 0.0% C 0.0% printf/fmt: L 0.0% C 0.0% startup/loader: L 100.0% C 0.0% compute: L 0.0% C 100.0%

(real stdout output is a write() syscall: 1-2 per whole run - see benchmarks/PROFILE.md; the purple above is in-memory number formatting)

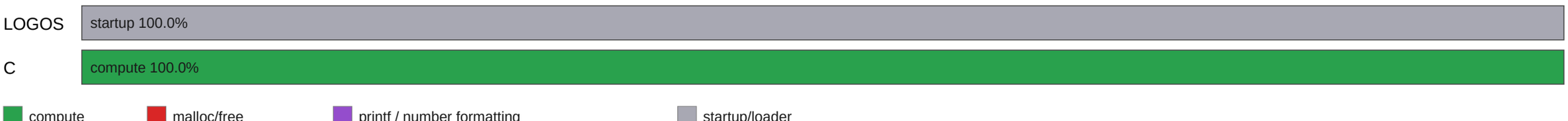


LOGOS compiler-optimizes this (memoized): finishes in <0.01s (vs C's full run) - nothing to profile, its whole runtime is process startup. Still 0% malloc, 0% printf.

ackermann - LOGOS vs C (n = 12)

malloc: L 0.0% C 0.0% printf/fmt: L 0.0% C 0.0% startup/loader: L 100.0% C 0.0% compute: L 0.0% C 100.0%

(real stdout output is a write() syscall: 1-2 per whole run - see benchmarks/PROFILE.md; the purple above is in-memory number formatting)



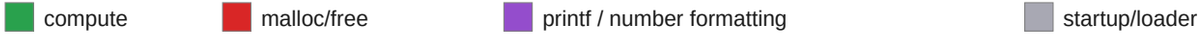
LOGOS compiler-optimizes this (memoized): finishes in <0.01s (vs C's full run) - nothing to profile, its whole runtime is process startup. Still 0% malloc, 0% printf.

nqueens - LOGOS vs C (n = 14)

malloc: L 0.0% C 0.0% printf/fmt: L 0.0% C 0.0% startup/loader: L 0.1% C 0.0% compute: L 99.9% C 100.0%

(real stdout output is a write() syscall: 1-2 per whole run - see benchmarks/PROFILE.md; the purple above is in-memory number formatting)





flamegraph = LOGOS on-CPU stacks (loader/thread startup trimmed); width = share of CPU time



bubble_sort - LOGOS vs C (n = 158,048)

malloc: L 0.0% C 0.0% printf/fmt: L 0.0% C 0.0% startup/loader: L 0.0% C 0.0% compute: L 100.0% C 100.0%

(real stdout output is a write() syscall: 1-2 per whole run - see benchmarks/PROFILE.md; the purple above is in-memory number formatting)



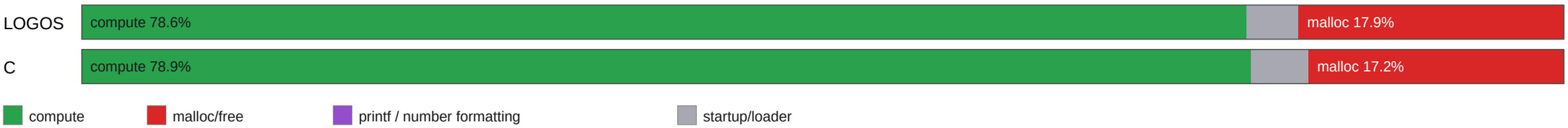
flamegraph = LOGOS on-CPU stacks (loader/thread startup trimmed); width = share of CPU time



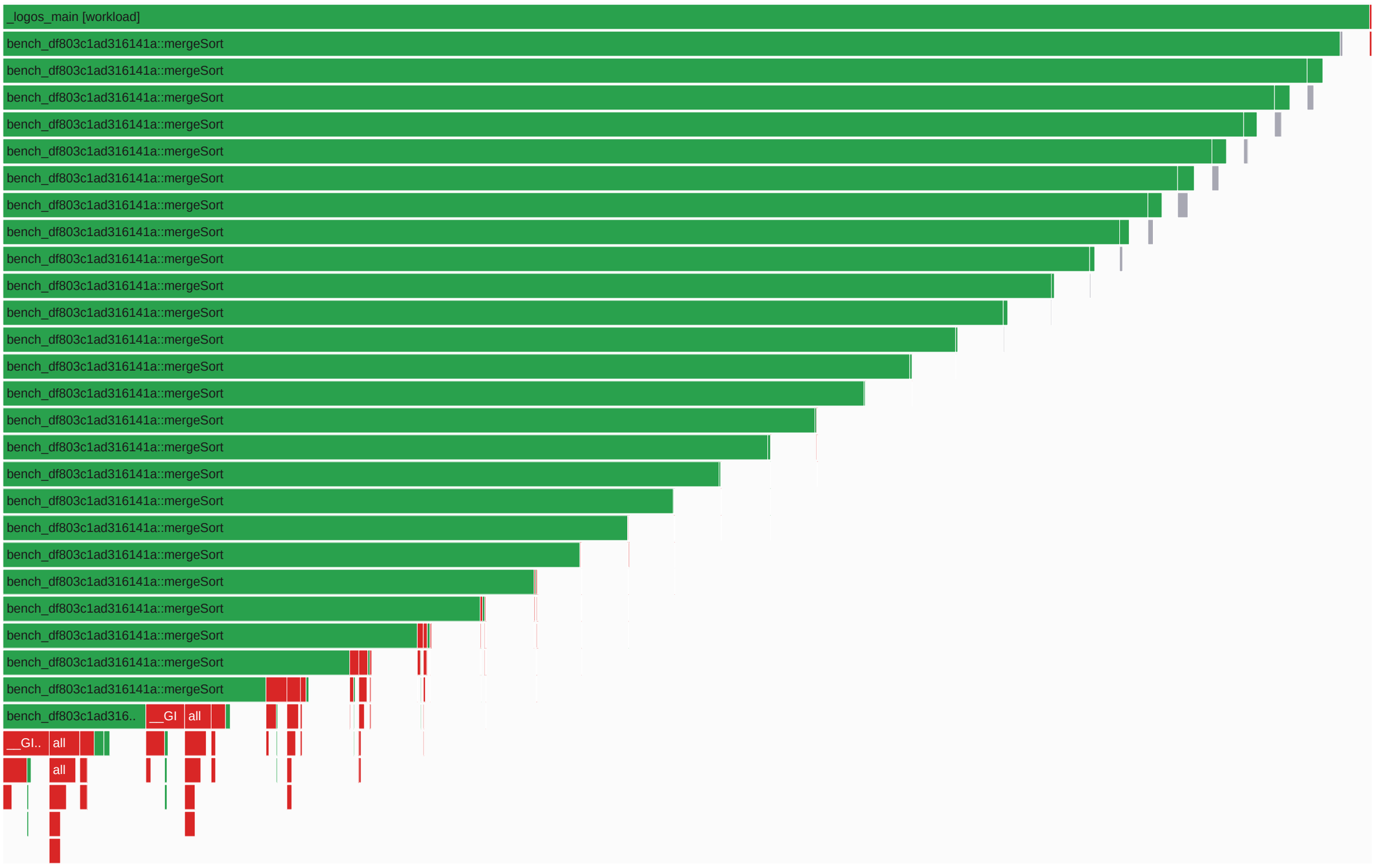
mergesort - LOGOS vs C (n = 55,386,315)

malloc: L 17.9% C 17.2% printf/fmt: L 0.0% C 0.0% startup/loader: L 3.5% C 3.9% compute: L 78.6% C 78.9%

(real stdout output is a write() syscall: 1-2 per whole run - see benchmarks/PROFILE.md; the purple above is in-memory number formatting)



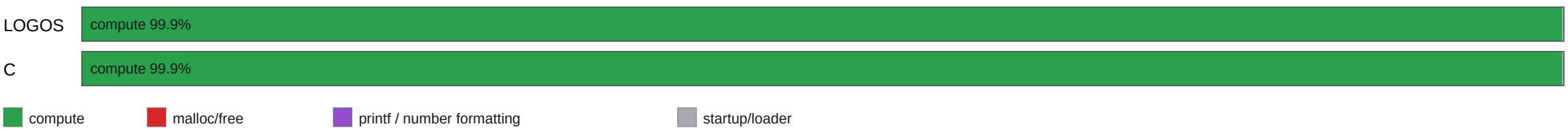
flamegraph = LOGOS on-CPU stacks (loader/thread startup trimmed); width = share of CPU time


_logos_main [workload]
bench_df803c1ad316141a::mergeSort
bench_df803c1ad316141a::mergeSort
bench_df803c1ad316141a::mergeSort

quicksort - LOGOS vs C (n = 52,276,512)

malloc: L 0.0% C 0.0% printf/fmt: L 0.0% C 0.0% startup/loader: L 0.1% C 0.1% compute: L 99.9% C 99.9%

(real stdout output is a write() syscall: 1-2 per whole run - see benchmarks/PROFILE.md; the purple above is in-memory number formatting)



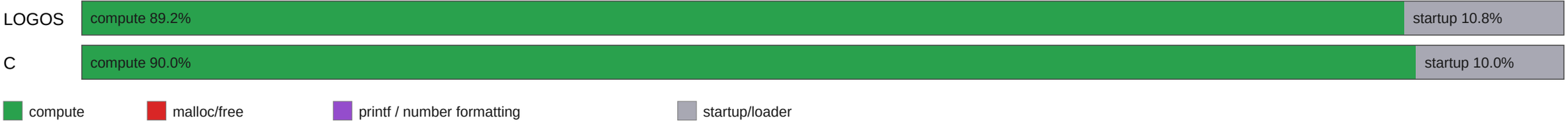
flamegraph = LOGOS on-CPU stacks (loader/thread startup trimmed); width = share of CPU time



counting_sort - LOGOS vs C (n = 565,125,712)

malloc: L 0.0% C 0.0% printf/fmt: L 0.0% C 0.0% startup/loader: L 10.8% C 10.0% compute: L 89.2% C 90.0%

(real stdout output is a write() syscall: 1-2 per whole run - see benchmarks/PROFILE.md; the purple above is in-memory number formatting)



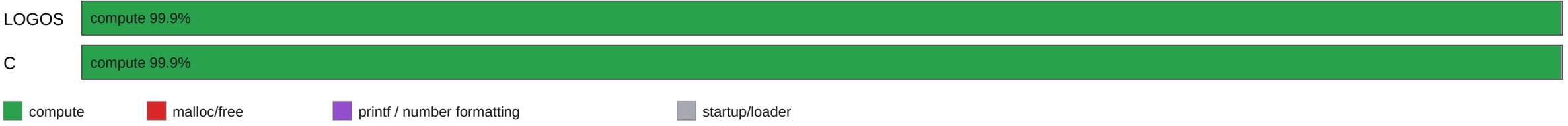
flamegraph = LOGOS on-CPU stacks (loader/thread startup trimmed); width = share of CPU time



heap_sort - LOGOS vs C (n = 48,241,020)

malloc: L 0.0% C 0.0% printf/fmt: L 0.0% C 0.0% startup/loader: L 0.1% C 0.1% compute: L 99.9% C 99.9%

(real stdout output is a write() syscall: 1-2 per whole run - see benchmarks/PROFILE.md; the purple above is in-memory number formatting)



flamegraph = LOGOS on-CPU stacks (loader/thread startup trimmed); width = share of CPU time



nbody - LOGOS vs C (n = 146,926,656)

malloc: L 0.0% C 0.0% printf/fmt: L 0.0% C 0.0% startup/loader: L 0.0% C 0.0% compute: L 100.0% C 100.0%

(real stdout output is a write() syscall: 1-2 per whole run - see benchmarks/PROFILE.md; the purple above is in-memory number formatting)



flamegraph = LOGOS on-CPU stacks (loader/thread startup trimmed); width = share of CPU time



mandelbrot - LOGOS vs C (n = 15,024)

malloc: L 0.0% C 0.0% printf/fmt: L 0.0% C 0.0% startup/loader: L 0.0% C 0.0% compute: L 100.0% C 100.0%

(real stdout output is a write() syscall: 1-2 per whole run - see benchmarks/PROFILE.md; the purple above is in-memory number formatting)



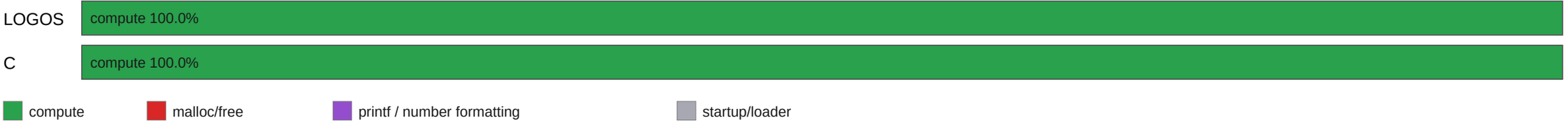
flamegraph = LOGOS on-CPU stacks (loader/thread startup trimmed); width = share of CPU time



spectral_norm - LOGOS vs C (n = 17,032)

malloc: L 0.0% C 0.0% printf/fmt: L 0.0% C 0.0% startup/loader: L 0.0% C 0.0% compute: L 100.0% C 100.0%

(real stdout output is a write() syscall: 1-2 per whole run - see benchmarks/PROFILE.md; the purple above is in-memory number formatting)



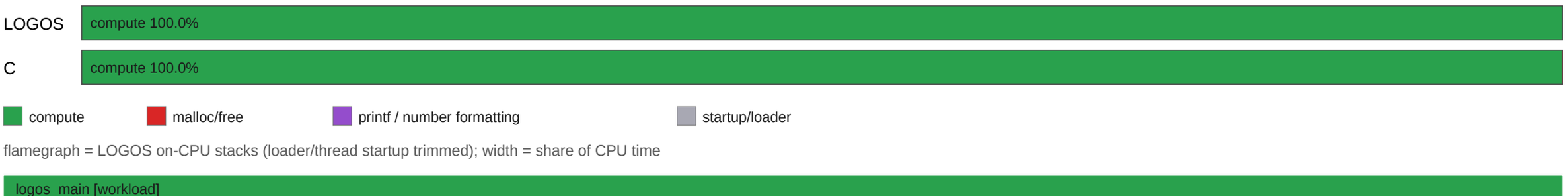
flamegraph = LOGOS on-CPU stacks (loader/thread startup trimmed); width = share of CPU time



pi_leibniz - LOGOS vs C (n = 5,833,224,944)

malloc: L 0.0% C 0.0% printf/fmt: L 0.0% C 0.0% startup/loader: L 0.0% C 0.0% compute: L 100.0% C 100.0%

(real stdout output is a write() syscall: 1-2 per whole run - see benchmarks/PROFILE.md; the purple above is in-memory number formatting)



flamegraph = LOGOS on-CPU stacks (loader/thread startup trimmed); width = share of CPU time

gcd - LOGOS vs C (n = 53,096)

malloc: L 0.0% C 0.0% printf/fmt: L 0.0% C 0.0% startup/loader: L 0.0% C 0.0% compute: L 100.0% C 100.0%

(real stdout output is a write() syscall: 1-2 per whole run - see benchmarks/PROFILE.md; the purple above is in-memory number formatting)



flamegraph = LOGOS on-CPU stacks (loader/thread startup trimmed); width = share of CPU time



collatz - LOGOS vs C (n = 29,177,848)

malloc: L 0.0% C 0.0% printf/fmt: L 0.0% C 0.0% startup/loader: L 0.0% C 0.0% compute: L 100.0% C 100.0%

(real stdout output is a write() syscall: 1-2 per whole run - see benchmarks/PROFILE.md; the purple above is in-memory number formatting)



flamegraph = LOGOS on-CPU stacks (loader/thread startup trimmed); width = share of CPU time



primes - LOGOS vs C (n = 22,704,512)

malloc: L 0.0% C 0.0% printf/fmt: L 0.0% C 0.0% startup/loader: L 0.0% C 0.0% compute: L 100.0% C 100.0%

(real stdout output is a write() syscall: 1-2 per whole run - see benchmarks/PROFILE.md; the purple above is in-memory number formatting)



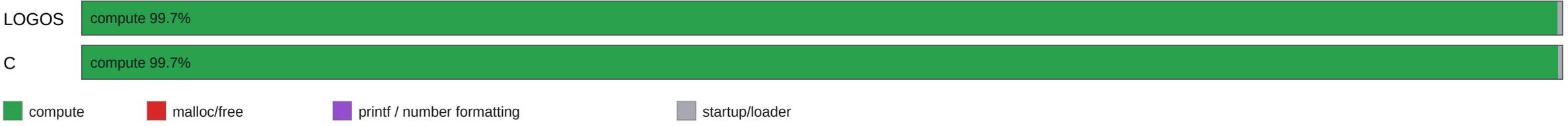
flamegraph = LOGOS on-CPU stacks (loader/thread startup trimmed); width = share of CPU time



sieve - LOGOS vs C (n = 697,912,470)

malloc: L 0.0% C 0.0% printf/fmt: L 0.0% C 0.0% startup/loader: L 0.3% C 0.3% compute: L 99.7% C 99.7%

(real stdout output is a write() syscall: 1-2 per whole run - see benchmarks/PROFILE.md; the purple above is in-memory number formatting)



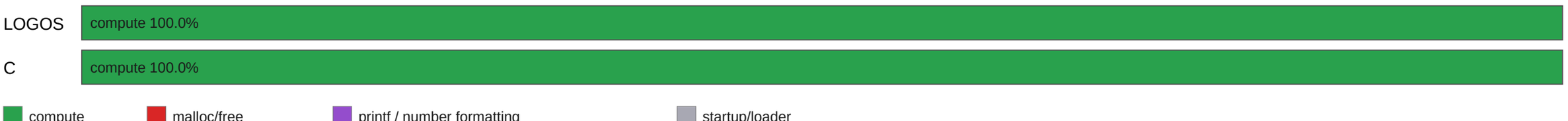
flamegraph = LOGOS on-CPU stacks (loader/thread startup trimmed); width = share of CPU time



matrix_mult - LOGOS vs C (n = 1,930)

malloc: L 0.0% C 0.0% printf/fmt: L 0.0% C 0.0% startup/loader: L 0.0% C 0.0% compute: L 100.0% C 100.0%

(real stdout output is a write() syscall: 1-2 per whole run - see benchmarks/PROFILE.md; the purple above is in-memory number formatting)



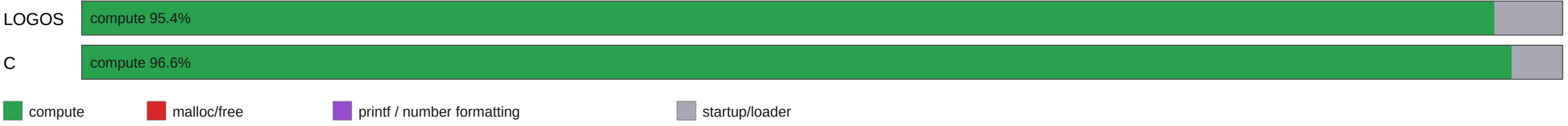
flamegraph = LOGOS on-CPU stacks (loader/thread startup trimmed); width = share of CPU time



prefix_sum - LOGOS vs C (n = 606,011,441)

malloc: L 0.0% C 0.0% printf/fmt: L 0.0% C 0.0% startup/loader: L 4.6% C 3.4% compute: L 95.4% C 96.6%

(real stdout output is a write() syscall: 1-2 per whole run - see benchmarks/PROFILE.md; the purple above is in-memory number formatting)



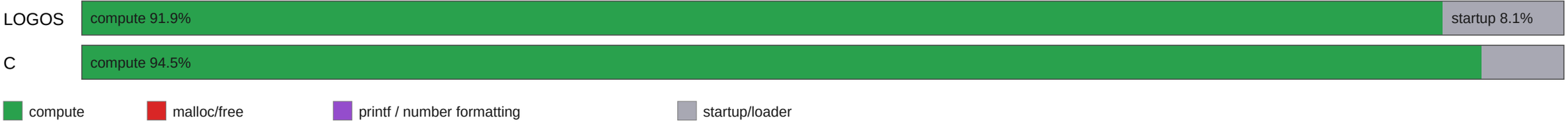
flamegraph = LOGOS on-CPU stacks (loader/thread startup trimmed); width = share of CPU time



array_reverse - LOGOS vs C (n = 854,578,669)

malloc: L 0.0% C 0.0% printf/fmt: L 0.0% C 0.0% startup/loader: L 8.1% C 5.5% compute: L 91.9% C 94.5%

(real stdout output is a write() syscall: 1-2 per whole run - see benchmarks/PROFILE.md; the purple above is in-memory number formatting)



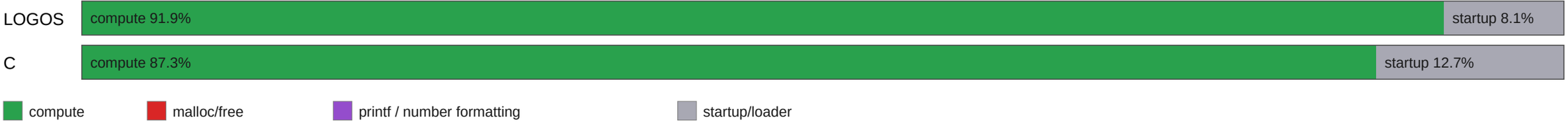
flamegraph = LOGOS on-CPU stacks (loader/thread startup trimmed); width = share of CPU time



array_fill - LOGOS vs C (n = 1,227,634,320)

malloc: L 0.0% C 0.0% printf/fmt: L 0.0% C 0.0% startup/loader: L 8.1% C 12.7% compute: L 91.9% C 87.3%

(real stdout output is a write() syscall: 1-2 per whole run - see benchmarks/PROFILE.md; the purple above is in-memory number formatting)



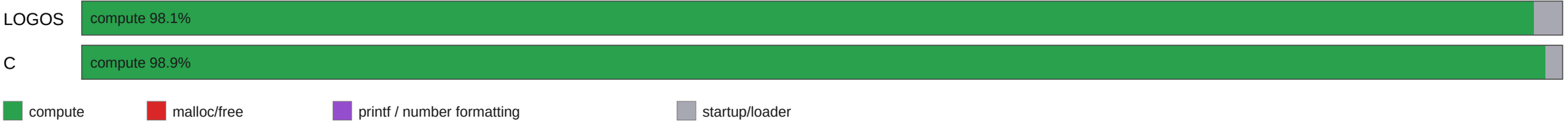
flamegraph = LOGOS on-CPU stacks (loader/thread startup trimmed); width = share of CPU time



collect - LOGOS vs C (n = 59,518,550)

malloc: L 0.0% C 0.0% printf/fmt: L 0.0% C 0.0% startup/loader: L 1.9% C 1.1% compute: L 98.1% C 98.9%

(real stdout output is a write() syscall: 1-2 per whole run - see benchmarks/PROFILE.md; the purple above is in-memory number formatting)



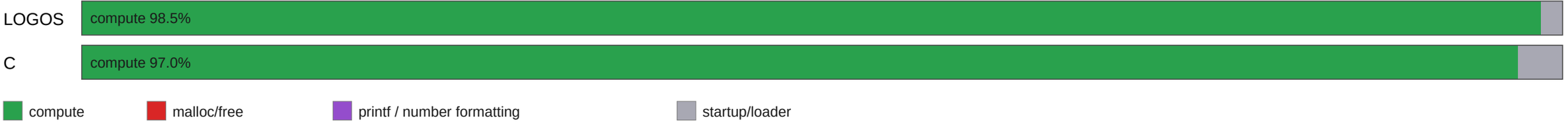
flamegraph = LOGOS on-CPU stacks (loader/thread startup trimmed); width = share of CPU time



two_sum - LOGOS vs C (n = 119,475,543)

malloc: L 0.0% C 0.0% printf/fmt: L 0.0% C 0.0% startup/loader: L 1.5% C 3.0% compute: L 98.5% C 97.0%

(real stdout output is a write() syscall: 1-2 per whole run - see benchmarks/PROFILE.md; the purple above is in-memory number formatting)



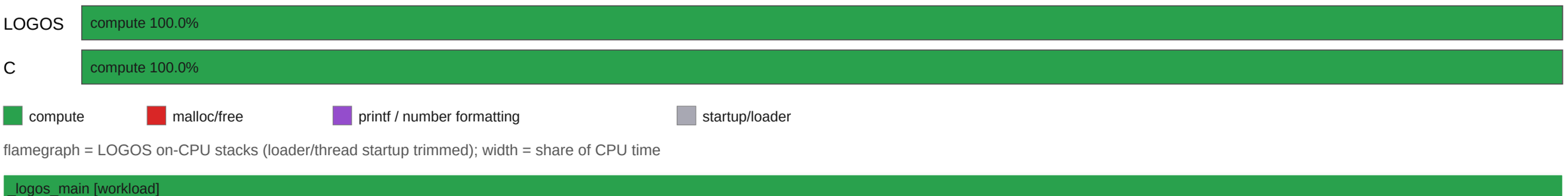
flamegraph = LOGOS on-CPU stacks (loader/thread startup trimmed); width = share of CPU time



histogram - LOGOS vs C (n = 2,386,643,440)

malloc: L 0.0% C 0.0% printf/fmt: L 0.0% C 0.0% startup/loader: L 0.0% C 0.0% compute: L 100.0% C 100.0%

(real stdout output is a write() syscall: 1-2 per whole run - see benchmarks/PROFILE.md; the purple above is in-memory number formatting)



flamegraph = LOGOS on-CPU stacks (loader/thread startup trimmed); width = share of CPU time

knapsack - LOGOS vs C (n = 73,136)

malloc: L 0.0% C 0.0% printf/fmt: L 0.0% C 0.0% startup/loader: L 0.0% C 0.0% compute: L 100.0% C 100.0%

(real stdout output is a write() syscall: 1-2 per whole run - see benchmarks/PROFILE.md; the purple above is in-memory number formatting)



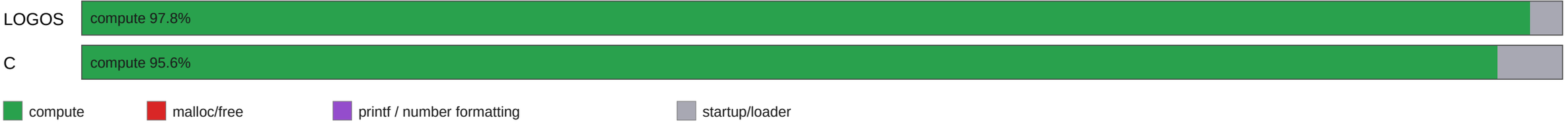
flamegraph = LOGOS on-CPU stacks (loader/thread startup trimmed); width = share of CPU time



coins - LOGOS vs C (n = 401,718,368)

malloc: L 0.0% C 0.0% printf/fmt: L 0.0% C 0.0% startup/loader: L 2.2% C 4.4% compute: L 97.8% C 95.6%

(real stdout output is a write() syscall: 1-2 per whole run - see benchmarks/PROFILE.md; the purple above is in-memory number formatting)



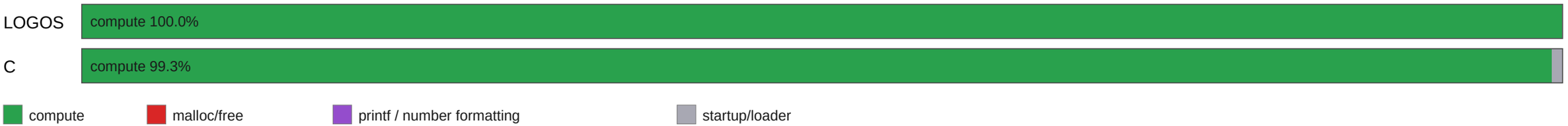
flamegraph = LOGOS on-CPU stacks (loader/thread startup trimmed); width = share of CPU time



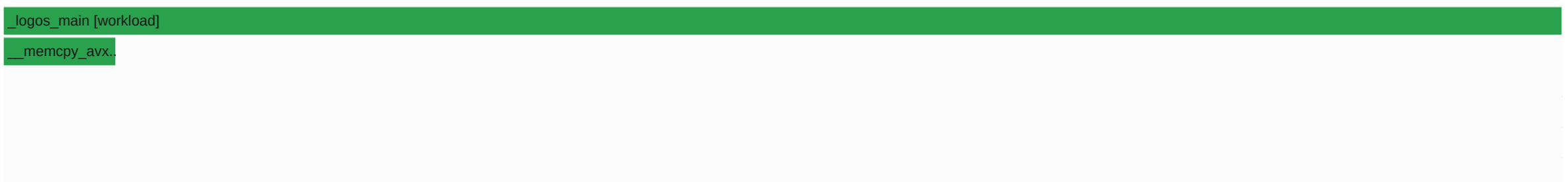
fannkuch - LOGOS vs C (n = 10)

malloc: L 0.0% C 0.0% printf/fmt: L 0.0% C 0.0% startup/loader: L 0.0% C 0.7% compute: L 100.0% C 99.3%

(real stdout output is a write() syscall: 1-2 per whole run - see benchmarks/PROFILE.md; the purple above is in-memory number formatting)



flamegraph = LOGOS on-CPU stacks (loader/thread startup trimmed); width = share of CPU time



strings - LOGOS vs C (n = 125,882,141)

malloc: L 0.0% C 0.0% printf/fmt: L 0.0% C 39.9% startup/loader: L 1.7% C 1.1% compute: L 98.2% C 59.0%

(real stdout output is a write() syscall: 1-2 per whole run - see benchmarks/PROFILE.md; the purple above is in-memory number formatting)



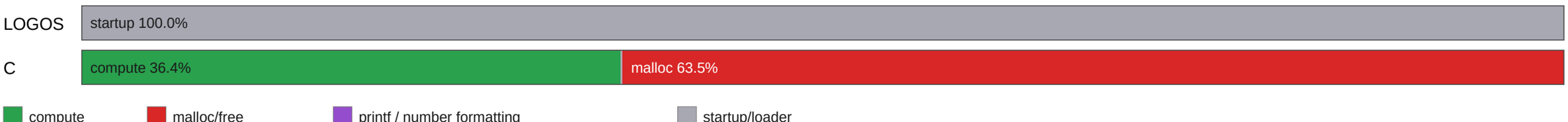
flamegraph = LOGOS on-CPU stacks (loader/thread startup trimmed); width = share of CPU time



binary_trees - LOGOS vs C (n = 23)

malloc: L 0.0% C 63.5% printf/fmt: L 0.0% C 0.0% startup/loader: L 100.0% C 0.1% compute: L 0.0% C 36.4%

(real stdout output is a write() syscall: 1-2 per whole run - see benchmarks/PROFILE.md; the purple above is in-memory number formatting)

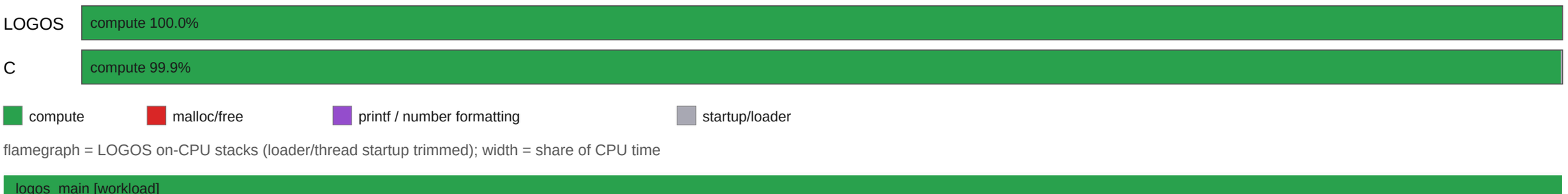


LOGOS compiler-optimizes this (closed-form): finishes in <0.01s (vs C's full run) - nothing to profile, its whole runtime is process startup. Still 0% malloc, 0% printf.

loop_sum - LOGOS vs C (n = 1,918,787,688)

malloc: L 0.0% C 0.0% printf/fmt: L 0.0% C 0.0% startup/loader: L 0.0% C 0.1% compute: L 100.0% C 99.9%

(real stdout output is a write() syscall: 1-2 per whole run - see benchmarks/PROFILE.md; the purple above is in-memory number formatting)

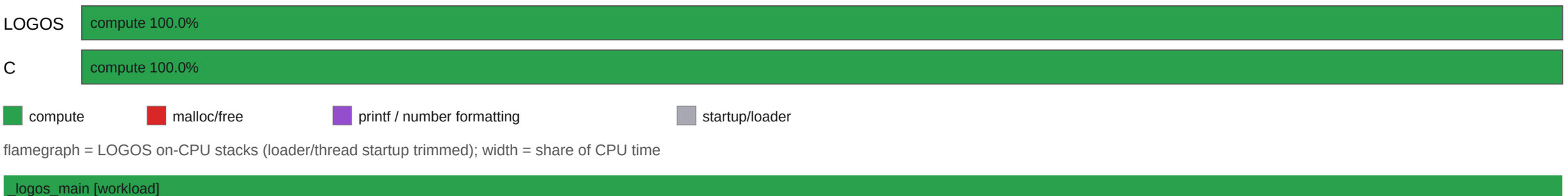


flamegraph = LOGOS on-CPU stacks (loader/thread startup trimmed); width = share of CPU time

fib_iterative - LOGOS vs C (n = 1,934,869,816)

malloc: L 0.0% C 0.0% printf/fmt: L 0.0% C 0.0% startup/loader: L 0.0% C 0.0% compute: L 100.0% C 100.0%

(real stdout output is a write() syscall: 1-2 per whole run - see benchmarks/PROFILE.md; the purple above is in-memory number formatting)

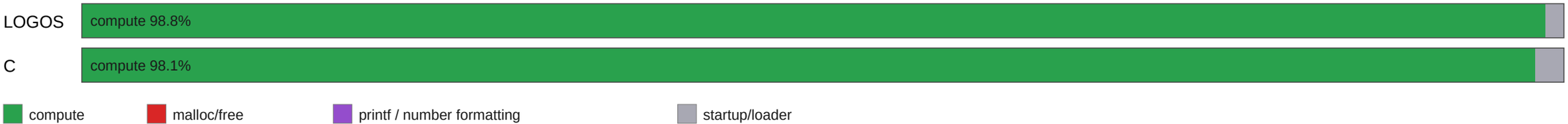


flamegraph = LOGOS on-CPU stacks (loader/thread startup trimmed); width = share of CPU time

graph_bfs - LOGOS vs C (n = 37,467,577)

malloc: L 0.0% C 0.0% printf/fmt: L 0.0% C 0.0% startup/loader: L 1.2% C 1.9% compute: L 98.8% C 98.1%

(real stdout output is a write() syscall: 1-2 per whole run - see benchmarks/PROFILE.md; the purple above is in-memory number formatting)



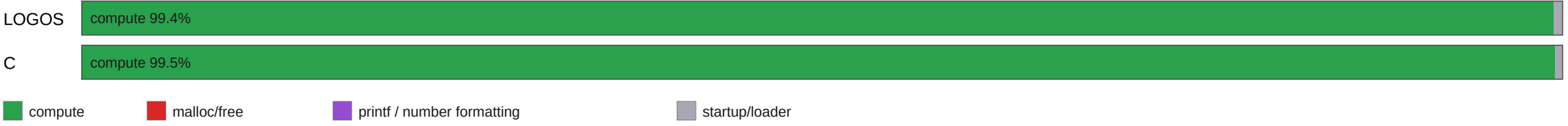
flamegraph = LOGOS on-CPU stacks (loader/thread startup trimmed); width = share of CPU time



string_search - LOGOS vs C (n = 924,964,604)

malloc: L 0.0% C 0.0% printf/fmt: L 0.0% C 0.0% startup/loader: L 0.6% C 0.5% compute: L 99.4% C 99.5%

(real stdout output is a write() syscall: 1-2 per whole run - see benchmarks/PROFILE.md; the purple above is in-memory number formatting)



flamegraph = LOGOS on-CPU stacks (loader/thread startup trimmed); width = share of CPU time



analytics - LOGOS vs C (n = 880,000,000)

malloc: L 0.0% C 0.0% printf/fmt: L 0.0% C 0.0% startup/loader: L 0.0% C 0.0% compute: L 100.0% C 100.0%

(real stdout output is a write() syscall: 1-2 per whole run - see benchmarks/PROFILE.md; the purple above is in-memory number formatting)



flamegraph = LOGOS on-CPU stacks (loader/thread startup trimmed); width = share of CPU time



vm - LOGOS vs C (n = 660,000,000)

malloc: L 0.0% C 0.0% printf/fmt: L 0.0% C 0.0% startup/loader: L 0.0% C 0.0% compute: L 100.0% C 100.0%

(real stdout output is a write() syscall: 1-2 per whole run - see benchmarks/PROFILE.md; the purple above is in-memory number formatting)



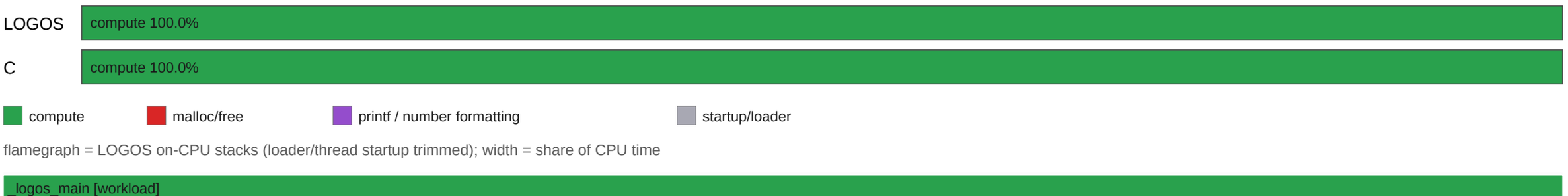
flamegraph = LOGOS on-CPU stacks (loader/thread startup trimmed); width = share of CPU time



sha256 - LOGOS vs C (n = 33,000,000)

malloc: L 0.0% C 0.0% printf/fmt: L 0.0% C 0.0% startup/loader: L 0.0% C 0.0% compute: L 100.0% C 100.0%

(real stdout output is a write() syscall: 1-2 per whole run - see benchmarks/PROFILE.md; the purple above is in-memory number formatting)



flamegraph = LOGOS on-CPU stacks (loader/thread startup trimmed); width = share of CPU time